

“Calculus 3”

Multi-Variable Calculus

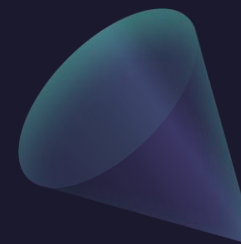
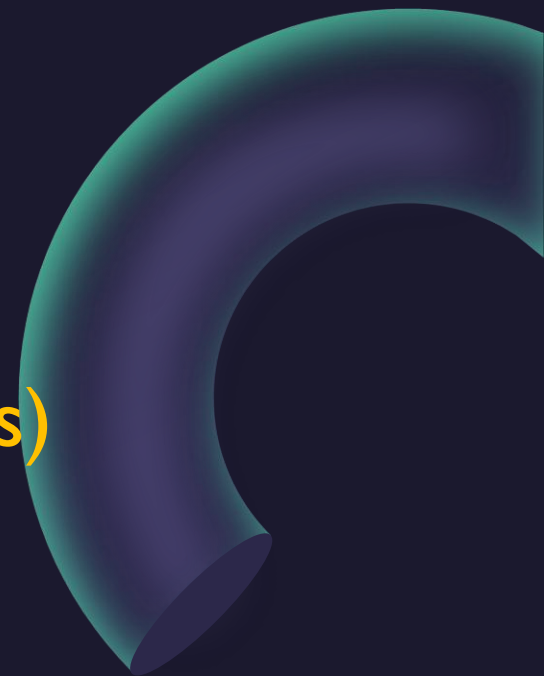
Instructor: Álvaro Lozano-Robledo

Day 20



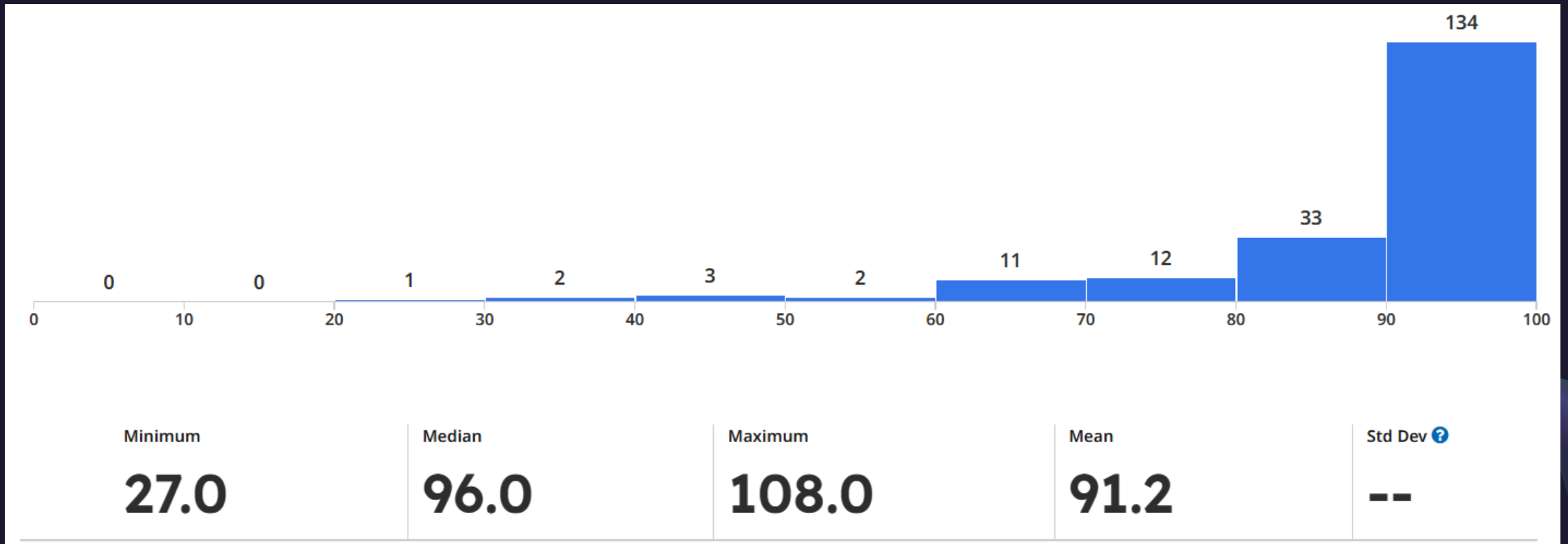
Any Reminders? Any Questions?

- Quiz on Friday (discussion) covers 16.2 (line integrals)
- Office hours Mondays 1-2pm online
- Office hours Thursday 3:30-4:30 at MONT 233
- Calc Night Thursday 6:30-8:30 at MONT 104



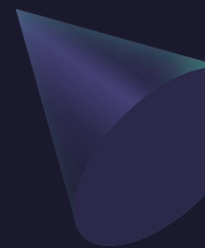


Exam 2: Stats!





ALVARO: Start the recording!

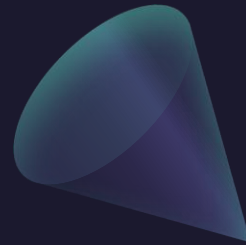
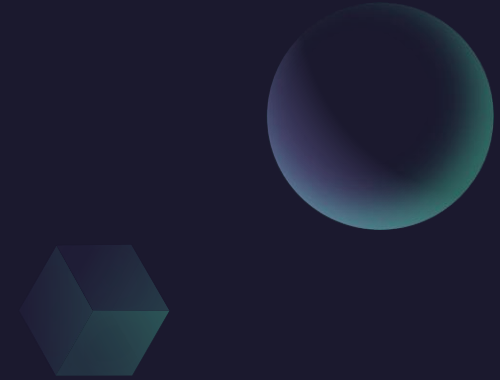


“Calculus 3”

Multi-Variable Calculus

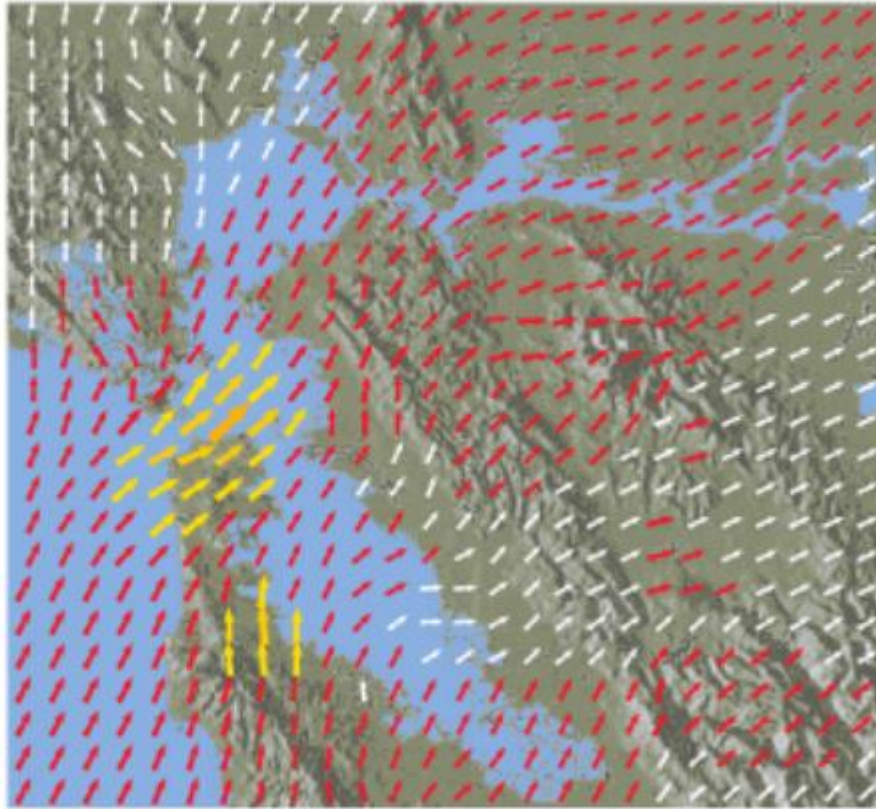
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Vector Fields

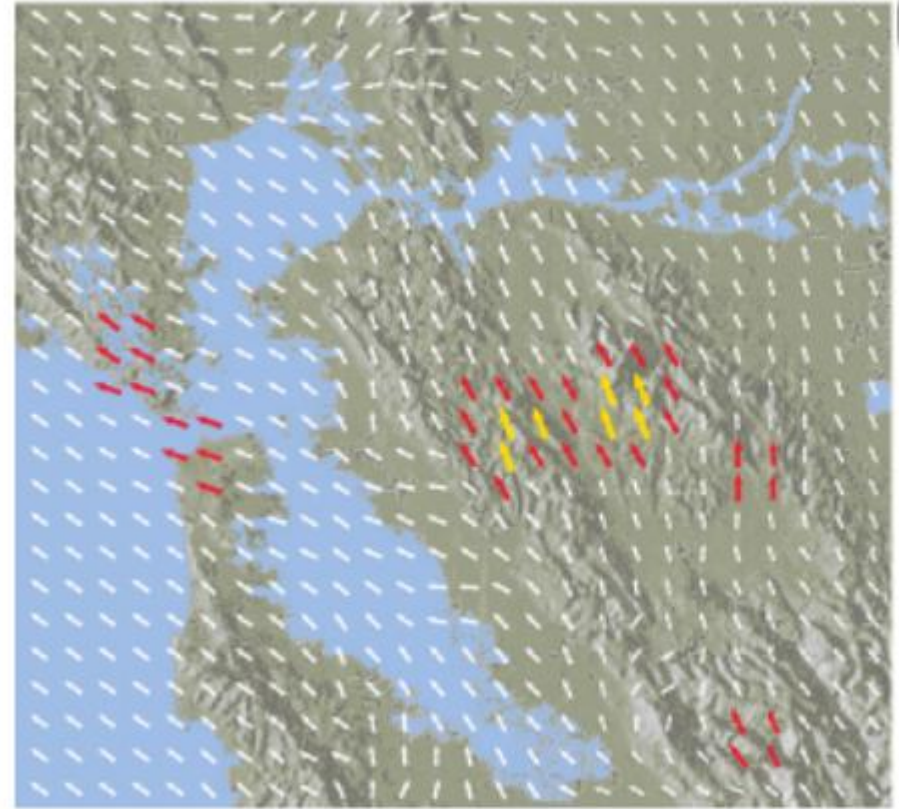


Vector Fields

Velocity vector fields showing San Francisco Bay wind patterns on a particular spring day



(a) 6:00 PM



(b) 6:00 AM

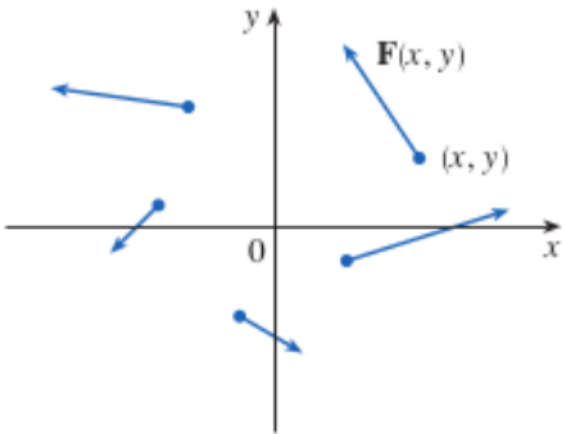
Vector Fields



Vector Fields

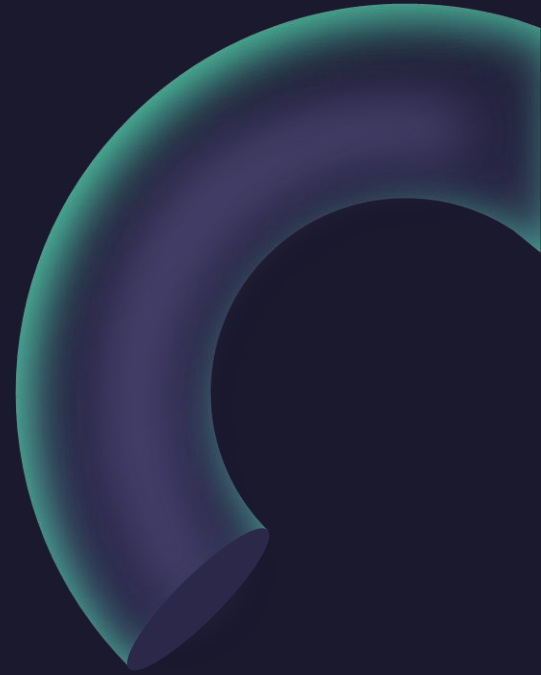
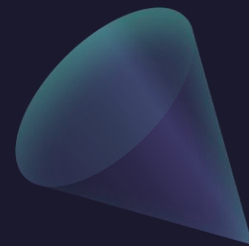
Let D be a set in $\mathbb{R}^2$ (a plane region). A **vector field on $\mathbb{R}^2$** is a function $\mathbf{F}$ that assigns to each point (x, y) in D a two-dimensional vector $\mathbf{F}(x, y)$.

Let E be a subset of $\mathbb{R}^3$. A **vector field on $\mathbb{R}^3$** is a function $\mathbf{F}$ that assigns to each point (x, y, z) in E a three-dimensional vector $\mathbf{F}(x, y, z)$.



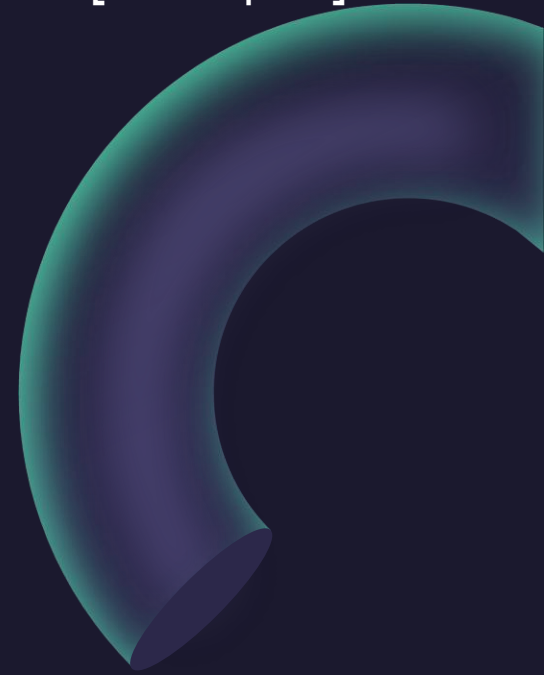
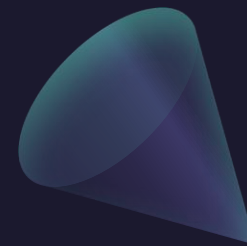
Example: Sketch a vector field on the plane given by

$$F(x, y) = (x, 0) = x \cdot i + 0 \cdot j$$



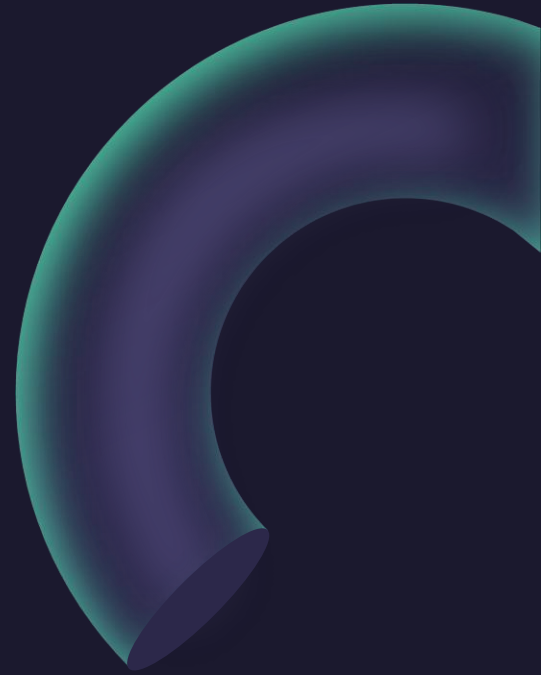
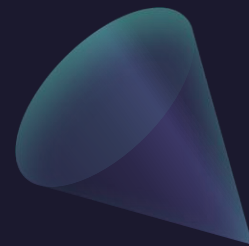
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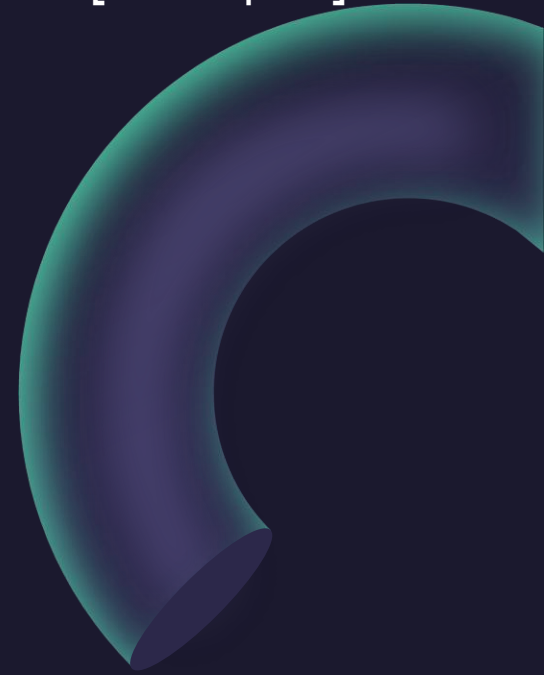
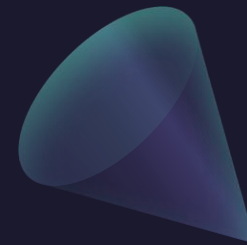


Example: Sketch a vector field on the plane given by

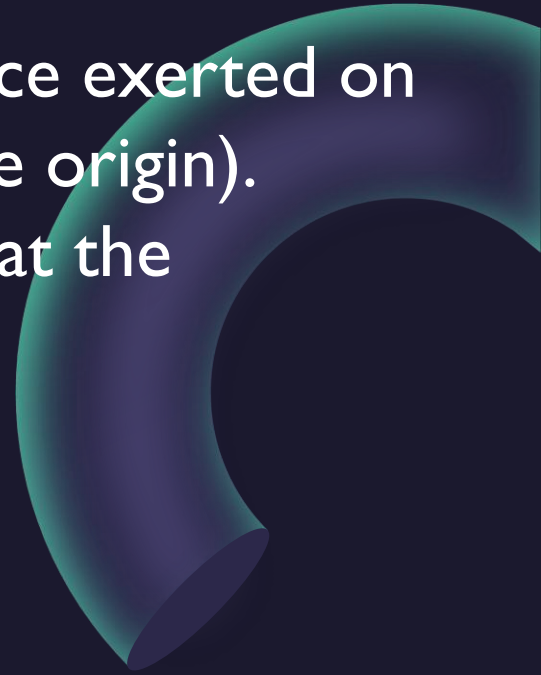
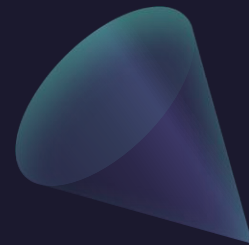
$$F(x, y) = (-y, x) = -y \cdot i + x \cdot j$$



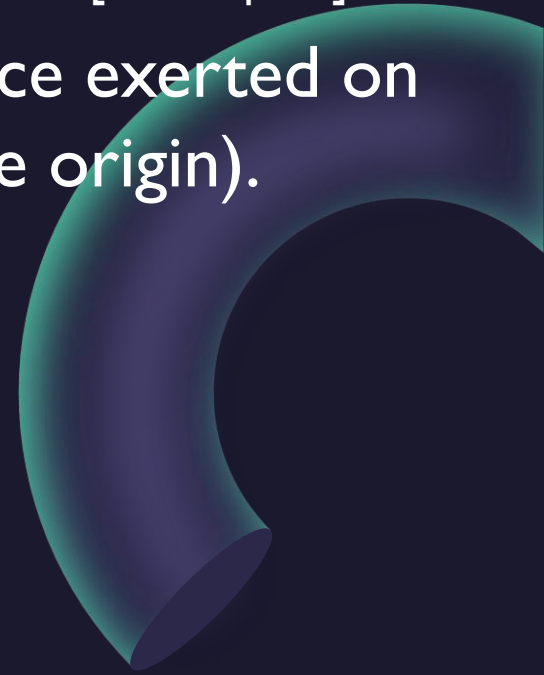
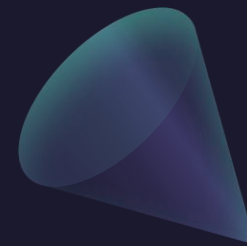
Example: Sketch a vector field on the plane given by

$$F(x, y) = (-y, x) = -y \cdot i + x \cdot j$$


Example: Sketch the gravitational vector field of the gravity force exerted on a planet of mass m by a star of mass M (located at the origin).
(Hint: recall that Newton's Laws of Gravitation say that the magnitude of the force is given by mMG/r^2 .)



Example: Sketch the gravitational vector field of the gravity force exerted on a planet of mass m by a star of mass M (located at the origin).



The Gradient Vector Field

If f is a function of two variables x and y , then the **gradient** of f is the vector function ∇f defined by

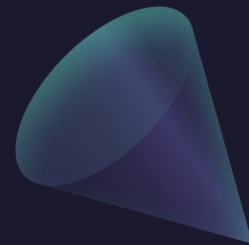
$$\nabla f(x, y) = \langle f_x(x, y), f_y(x, y) \rangle = \frac{\partial f}{\partial x} \mathbf{i} + \frac{\partial f}{\partial y} \mathbf{j}$$

$$\nabla f(x, y) = f_x(x, y) \mathbf{i} + f_y(x, y) \mathbf{j}$$

$$\nabla f(x, y, z) = f_x(x, y, z) \mathbf{i} + f_y(x, y, z) \mathbf{j} + f_z(x, y, z) \mathbf{k}$$

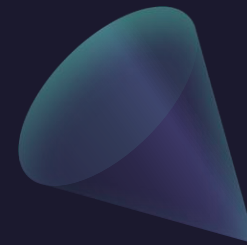
Example: Sketch the gradient vector field on the plane for the function

$$f(x, y) = x^2 + y^2$$

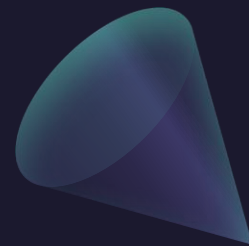
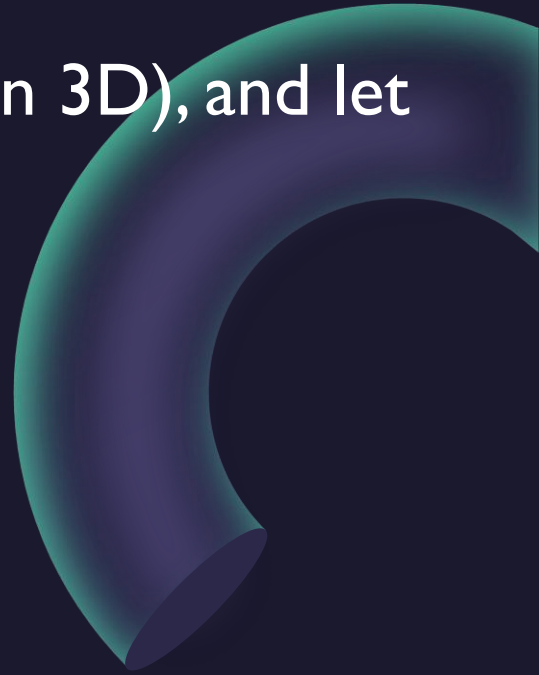


Example: Sketch the gradient vector field on the plane for the function

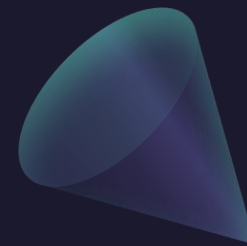
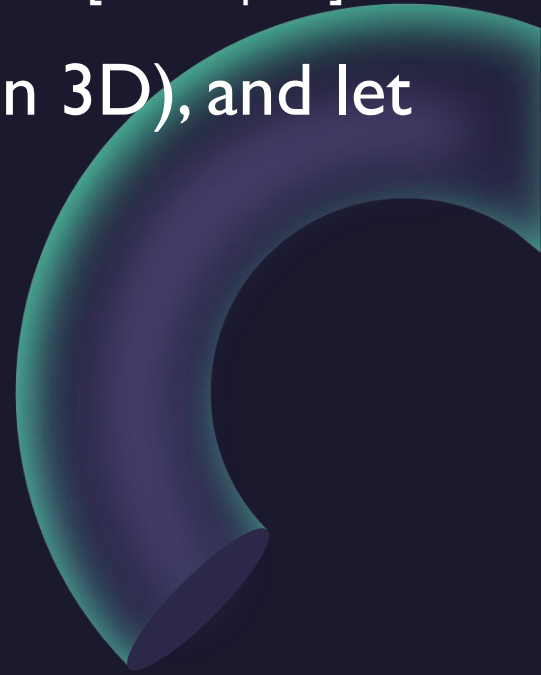
$$f(x, y) = x^2 + y^2$$



Example: Let $F(\vec{x}) = -\frac{mMG}{|\vec{x}|^3} \vec{x}$ be a gravitational vector field (in 3D), and let $f(\vec{x}) = \frac{mMG}{|\vec{x}|}$. Show that $F = \nabla f$.



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The Gradient Vector Field

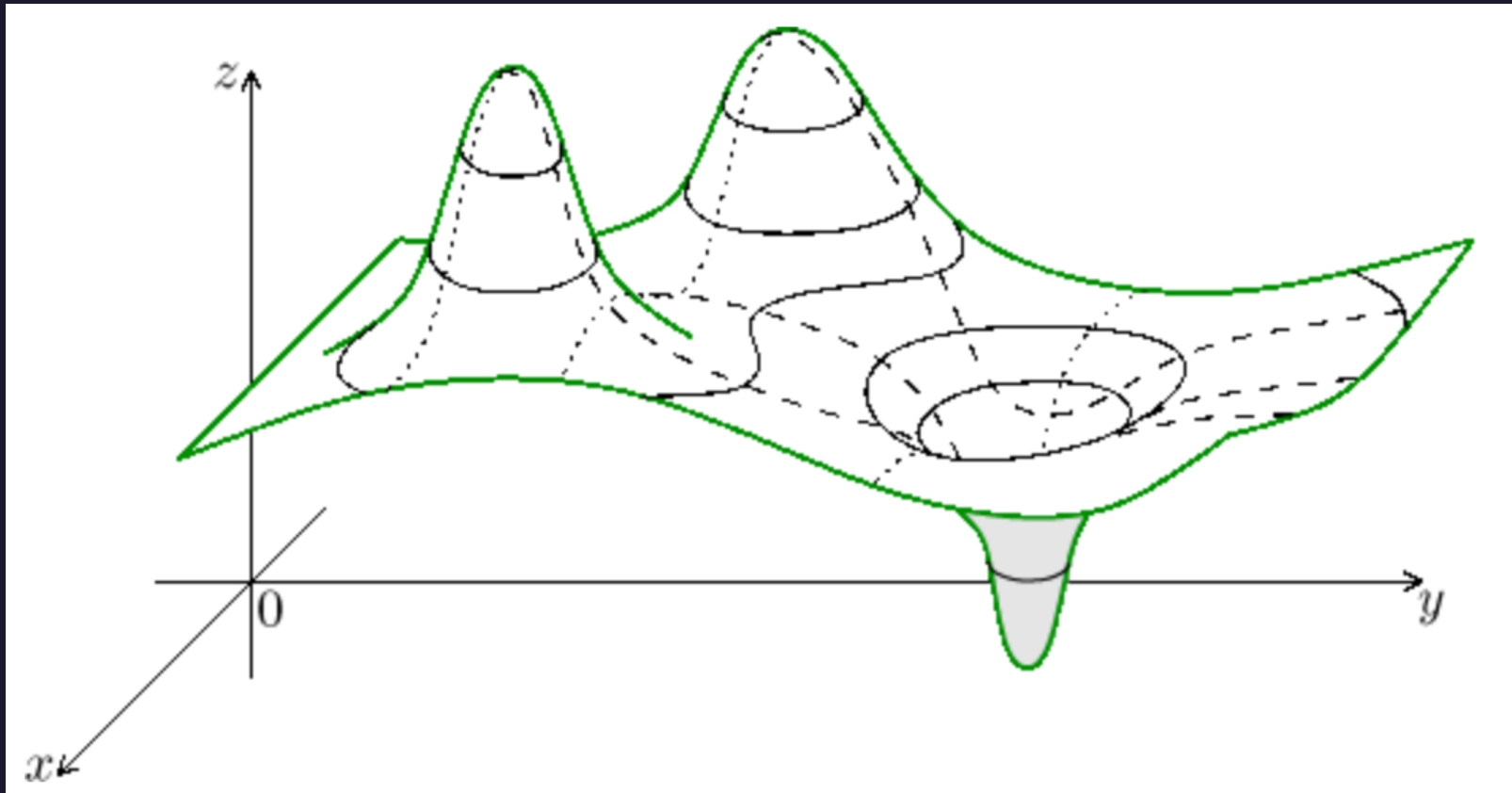
If f is a function of two variables x and y , then the **gradient** of f is the vector function ∇f defined by

$$\nabla f(x, y) = \langle f_x(x, y), f_y(x, y) \rangle = \frac{\partial f}{\partial x} \mathbf{i} + \frac{\partial f}{\partial y} \mathbf{j}$$

15 Theorem

Suppose f is a differentiable function of two or three variables. The maximum value of the directional derivative $D_{\mathbf{u}} f(\mathbf{x})$ is $|\nabla f(\mathbf{x})|$ and it occurs when $\mathbf{u}$ has the same direction as the gradient vector $\nabla f(\mathbf{x})$.

The Gradient Vector Field and “Gradient Descent” on “loss functions” in the training of AI/LLM models



Line Integrals of Vector Fields: An Example

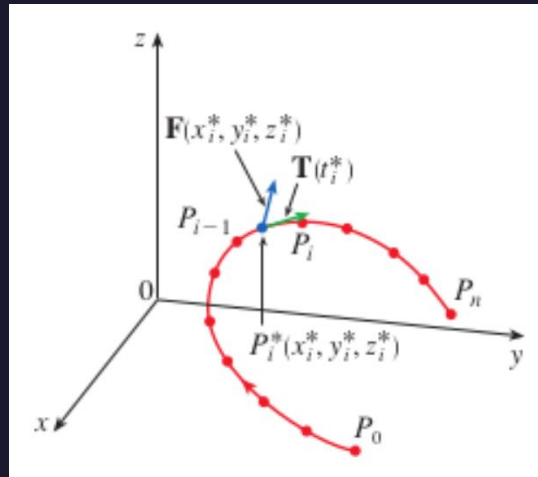
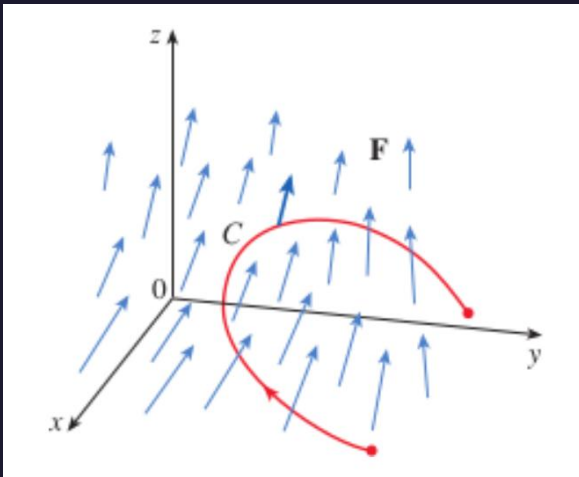
Recall that the “work” done by a constant force F (in vector form) in moving an object from point P to point Q in space is given by

$$W = F \cdot D$$

where $D = \overrightarrow{PQ}$.

Line Integrals of Vector Fields: An Example

What is the “work” done by a force $\mathbf{F}$ (in vector form) in moving an object from point P to point Q in space as we travel over a curve $\mathbf{r}(t)$?

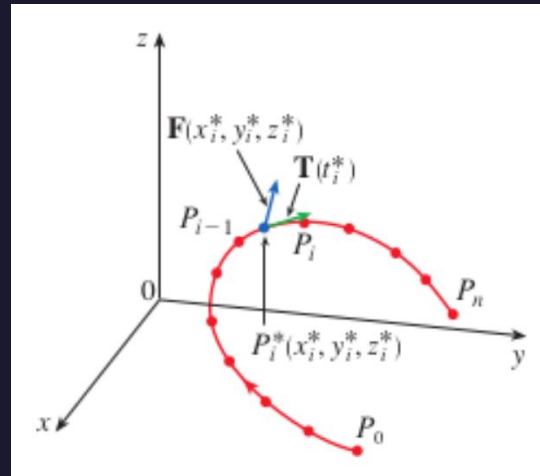
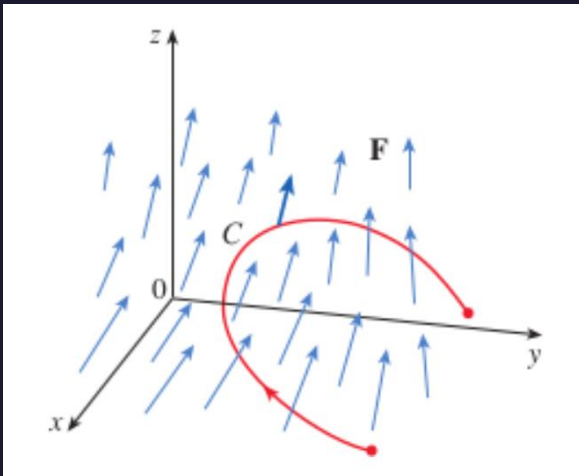


$$\sum_{i=1}^n [\mathbf{F}(x_i^*, y_i^*, z_i^*) \cdot \mathbf{T}(x_i^*, y_i^*, z_i^*)] \Delta s_i$$

$$W = \int_C \mathbf{F}(x, y, z) \cdot \mathbf{T}(x, y, z) ds = \int_C \mathbf{F} \cdot \mathbf{T} ds$$

Line Integrals of Vector Fields: An Example

What is the “work” done by a force $\mathbf{F}$ (in vector form) in moving an object from point P to point Q in space as we travel over a curve $\mathbf{r}(t)$?



$$\sum_{i=1}^n [\mathbf{F}(x_i^*, y_i^*, z_i^*) \cdot \mathbf{T}(x_i^*, y_i^*, z_i^*)] \Delta s_i$$

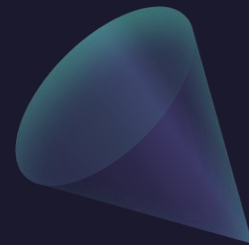
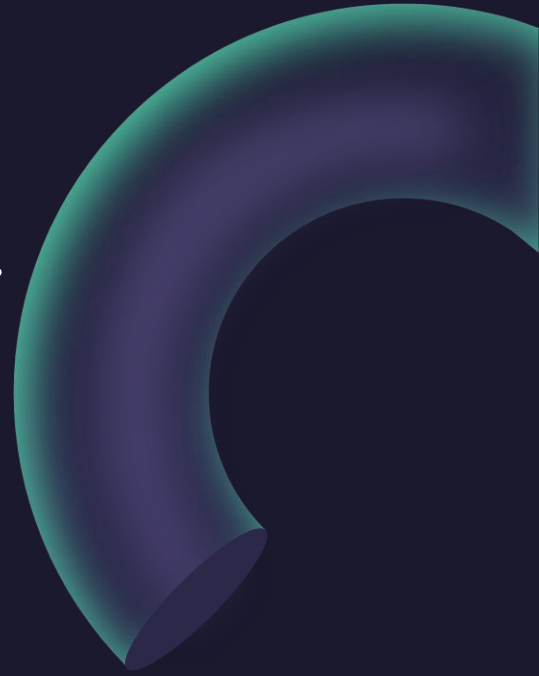
$$W = \int_C \mathbf{F}(x, y, z) \cdot \mathbf{T}(x, y, z) ds = \int_C \mathbf{F} \cdot \mathbf{T} ds$$

$$W = \int_a^b \left[\mathbf{F}(\mathbf{r}(t)) \cdot \frac{\mathbf{r}'(t)}{|\mathbf{r}'(t)|} \right] |\mathbf{r}'(t)| dt = \int_a^b \mathbf{F}(\mathbf{r}(t)) \cdot \mathbf{r}'(t) dt$$

Example: Find the work done by the force

$$F(x, y) = x^2 \cdot i - xy \cdot j$$

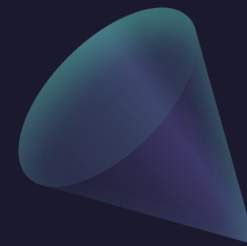
in moving a particle along the quarter circle from (1,0) to (0,1).



Example: Find the work done by the force

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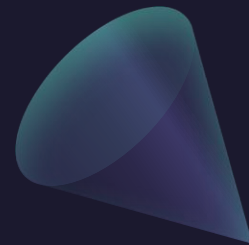


Line Integrals of Vector Fields: An Example

“Work” represents the amount of energy necessary to move an object in a particular path...



Example: Set up an integral that describes the work done by gravitational forces on Artemis II during its path $\mathbf{r}(t)$.



Line Integrals of Vector Fields in General

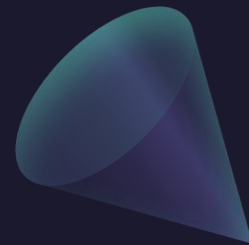
Let $\mathbf{F}$ be a continuous vector field defined on a smooth curve C given by a vector function $\mathbf{r}(t)$, $a \leq t \leq b$. Then the **line integral of $\mathbf{F}$ along C** is

$$\int_C \mathbf{F} \cdot d\mathbf{r} = \int_a^b \mathbf{F}(\mathbf{r}(t)) \cdot \mathbf{r}'(t) dt = \int_C \mathbf{F} \cdot \mathbf{T} ds$$

Example: Evaluate $\int_C F \cdot dr$ where

$$F(x, y, z) = xy \cdot i + yz \cdot j + xz \cdot k$$

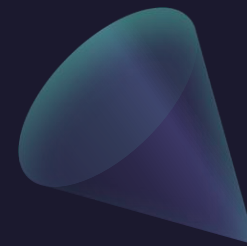
and C is the “twisted cubic” given by $r(t) = (t, t^2, t^3)$, $0 \leq t \leq 1$.



Example: Evaluate $\int_C F \cdot dr$ where

$$F(x, y, z) = xy \cdot i + yz \cdot j + xz \cdot k$$

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Line Integrals of Vector Fields in General

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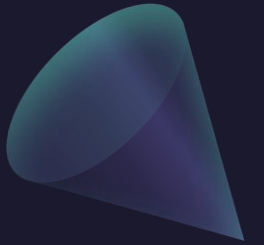
$$\int_C \mathbf{F} \cdot d\mathbf{r} = \int_a^b \mathbf{F}(\mathbf{r}(t)) \cdot \mathbf{r}'(t) dt = \int_C \mathbf{F} \cdot \mathbf{T} ds$$

$$\int_C \mathbf{F} \cdot d\mathbf{r} = \int_C P dx + Q dy + R dz \quad \text{where } \mathbf{F} = P \mathbf{i} + Q \mathbf{j} + R \mathbf{k}$$

Line Integrals of Vector Fields in General

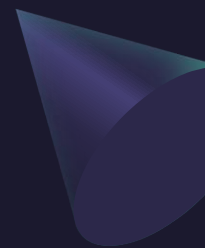
$$\int_C \mathbf{F} \cdot d\mathbf{r} = \int_C P \, dx + Q \, dy + R \, dz \quad \text{where } \mathbf{F} = P \mathbf{i} + Q \mathbf{j} + R \mathbf{k}$$

Questions?

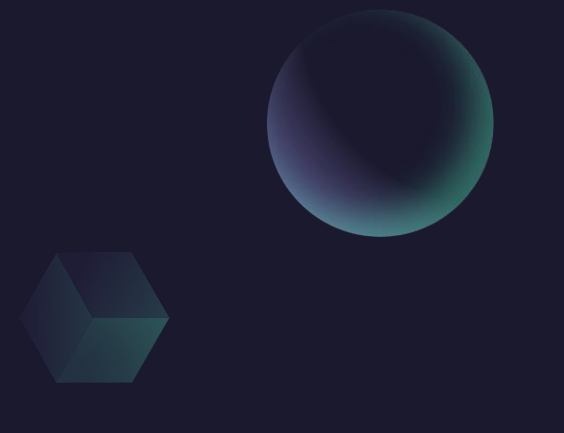




ALVARO: Start the recording!



“Calculus 3”

A sphere, a cube, and a cone are positioned in the upper right area of the slide. The sphere is at the top right, the cube is below it, and the cone is further down and to the left.

Multi-Variable Calculus

Instructor: Álvaro Lozano-Robledo

The Fundamental Theorem for Line Integrals

A cone and a torus are positioned in the lower right area of the slide. The cone is located between the 'Multi-Variable Calculus' and 'The Fundamental Theorem' text, and the torus is on the far right side of the slide.

The Fundamental Theorem of Calculus

The Fundamental Theorem of Calculus, Part 2

If f is continuous on $[a, b]$, then

$$\int_a^b f(x) \, dx = F(b) - F(a)$$

where F is any antiderivative of f , that is, a function F such that $F' = f$.

$$\int_a^b F'(x) \, dx = F(b) - F(a)$$

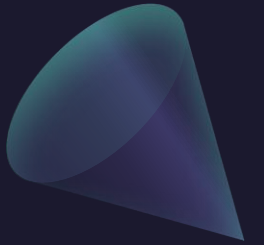
The Fundamental Theorem for Line Integrals

2 Theorem

Let C be a smooth curve given by the vector function $\mathbf{r}(t)$, $a \leq t \leq b$. Let f be a differentiable function of two or three variables whose gradient vector ∇f is continuous on C . Then

$$\int_C \nabla f \cdot d\mathbf{r} = f(\mathbf{r}(b)) - f(\mathbf{r}(a))$$

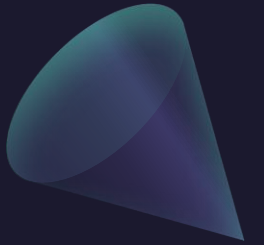
The Fundamental Theorem for Line Integrals: Proof



The Fundamental Theorem for Line Integrals: Proof

$$\begin{aligned}\int_C \nabla f \cdot d\mathbf{r} &= \int_a^b \nabla f(\mathbf{r}(t)) \cdot \mathbf{r}'(t) dt \\&= \int_a^b \left(\frac{\partial f}{\partial x} \frac{dx}{dt} + \frac{\partial f}{\partial y} \frac{dy}{dt} + \frac{\partial f}{\partial z} \frac{dz}{dt} \right) dt \\&= \int_a^b \frac{d}{dt} f(\mathbf{r}(t)) dt \quad \text{(by the Chain Rule)} \\&= f(\mathbf{r}(b)) - f(\mathbf{r}(a))\end{aligned}$$

Questions?



Thank you

Until next time.

